Applications of Java:

Java is used to develop various types of applications:

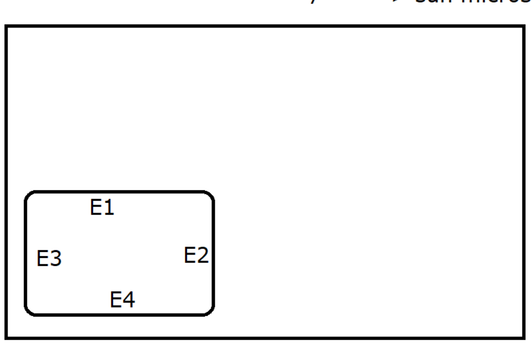
a) Desktop OR Standalone Applications (e.g., IDEs, tools)
b) Web Applications
c) Mobile Applications (Android)
d) Enterprise Applications (banking, ERP, etc.)
e) Gaming Applications
f) Cloud-based and AI/ML Applications
g) Distributed Application

Editions of Java:
----
a) Java SE (Java Standard Edition) Standalone/Desktop Apps
b) Java EE (Java Enterprise Edition) Web & Enterprise Apps
c) Java ME (Java Micro Edition)   Mobile & Embedded Devices
d) JavaFX (Java FX) Rich Internet Apps (UI/Graphics) [Outdated]

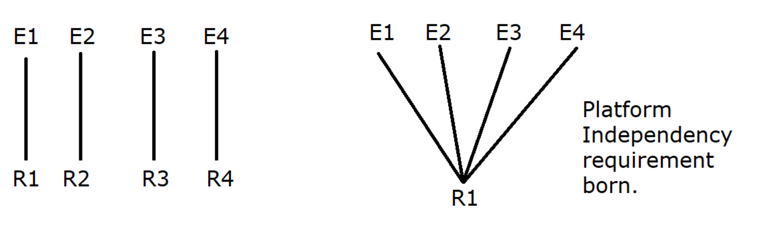
History of Java

Name of the language : Java
Initial Name of Java : OAK (Tree Name)
Developed By : James Gosling & his friends
Project Name : Green Project
Purpose : To create a common remote control
Java : Island in Indonesia
Official Symbol : Coffee CUP

Standford University   -> sun microsystem

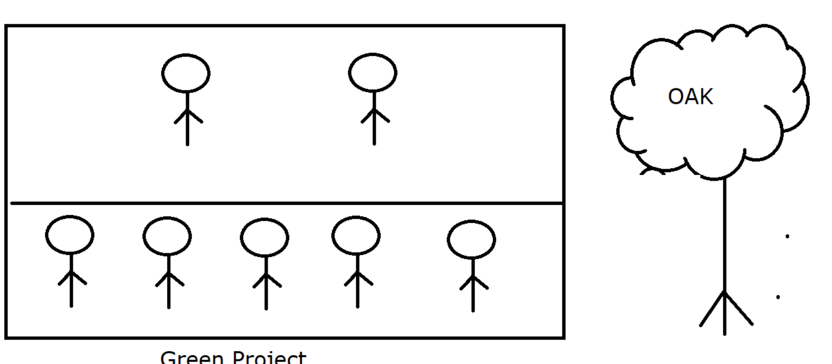


Can we create a common remote control ??



C and C++ -> Failed (Platform Dependent)

In this existing language, we need to enhance something to achieve this particular task



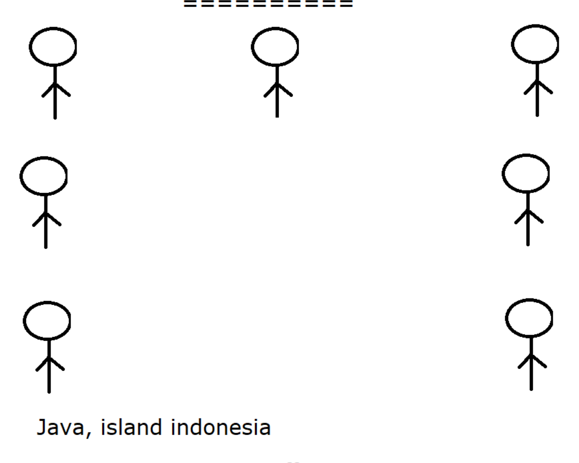
Green Project

Star 7 :

OAK (Tree Name)

Sent this name to Lab for Registration purpose

Rename of OAK
==========

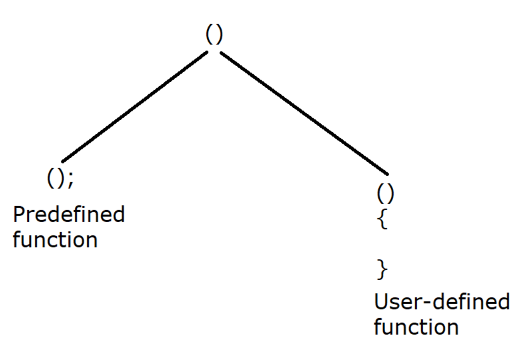


Java, island indonesia

JAVA -> Coffee CUP

What is a function?

() -> Symbol of function



* A function is a self defined block which is used to writing **business logic**, some calculation OR printing the result.

* A function may OR may not return a value.

Case 1 :

A function which is not returning a value :

```
public void accept()
{

}
```

Case 2 :

A function which is returning an int value :

```
public int accept()
{
  return 9;
}
```

A function is divided into two types:

1) Predefined OR Built-in function
2) User-defined OR Custom function

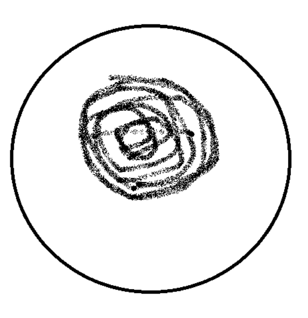
Predefined OR Built-in function

A function which is developed by language creator itself is called predefined function.

User-defined OR Custom Function

A function which is written by user/developer for its own requirement and specification is called User-defined function.

Advantages of a function :



Approach 1:

```
void main()
{
  int sum = a + b;

  int sub = a - b;

 int mul = a * b;

 int div = a/b;

}
```

1) Modularity

2) Easy to understand

3) Reusability

4) Easy Debugging

Approach 2:

```
void main()
{
  int sum(int x , int y)
  {
    return x + y;
  }

  int sub(int x , int y)
  {
    return x - y;
  }

  int mul(int x , int y)
  {
    return x * y;
  }

  int div(int x , int y)
  {
    return x / y;
  }
}
```